# 13D research: ML, NLP and long-term active risk

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**Investment aim:** use machine learning (ML) and natural language processing (NLP) to uncover long-term alpha signals in public companies backed by high-quality event-driven or activist hedge funds, then take active risk in several selected, potentially concentrated positions.

**Status:** the historical simple-following baseline has not established a reliable investable edge. Filing preparation and manager scorecards are underway; predictive ML and incremental intent NLP have not been evaluated.

The intended output is a ranked set of company opportunities with an economic thesis, catalyst, downside case and position rationale. Manager reputation and a disclosed plan are candidate inputs; their value must be tested using information available before purchase.

## 1. Three phases - baseline, selection and disclosed intentions

| Phase / purpose | Information and comparison | What has been done |
|---|---|---|
| **1. Simple following**<br>Measure post-disclosure returns. | Initial 13D events and manager groups; stock returns versus Russell size-segment ETF proxies. | Historical baseline evaluated. No reliable edge established; some funded simulations incomplete. |
| **2. Structured ML**<br>Rank stronger opportunities. | Investor history, ownership, company size, liquidity, volatility and prior returns; fundamentals planned. | 15,813 candidates audited; HFR scorecards prepared. Dated joins/features unfinished; no fitted model. |
| **3. Incremental NLP**<br>Add the investor's stated plan. | Item 4 objectives and specificity; compare structured ML with ML plus text on the same sample. | Planned. No validated intent/alpha result. Ownership/identity extraction is preparation work. |

## 2. Data sources and their role

| Source | Information | Use / status |
|---|---|---|
| **SEC EDGAR** | 13D/13D-A documents, public acceptance times, filer/issuer identities, stakes and Item 4 text. | Cached source-audited filing archive; factual extraction and future intent NLP. [E2] |
| **CRSP via WRDS** | Daily stock/ETF prices, returns, volume, historical security IDs and delisting fields. | Market data behind the reported historical return tests and simulations. [E1, E4] |
| **FTSE Russell via WRDS** | Historical Russell constituent snapshots; ETF returns come from CRSP. | Stock-appropriate size-segment ETF benchmark assignments. [E4] |
| **HFR** | 35 supplied quarterly snapshots, 2018 Q1-2026 Q3: fund metadata, strategy, reported returns and assets. | Manager candidates and descriptive peer scorecards; predictive value untested. [E3] |
| **Compustat via WRDS** | Fundamentals for valuation, leverage, profitability and balance-sheet features. | Planned features. A historically available, joined ML panel is unfinished. [E4] |

**WRDS** = Wharton Research Data Services, the access platform. **Yahoo Finance** supplied an early ten-company price-only preview, separate from these CRSP-based results; the preview script used Yahoo directly rather than yfinance.

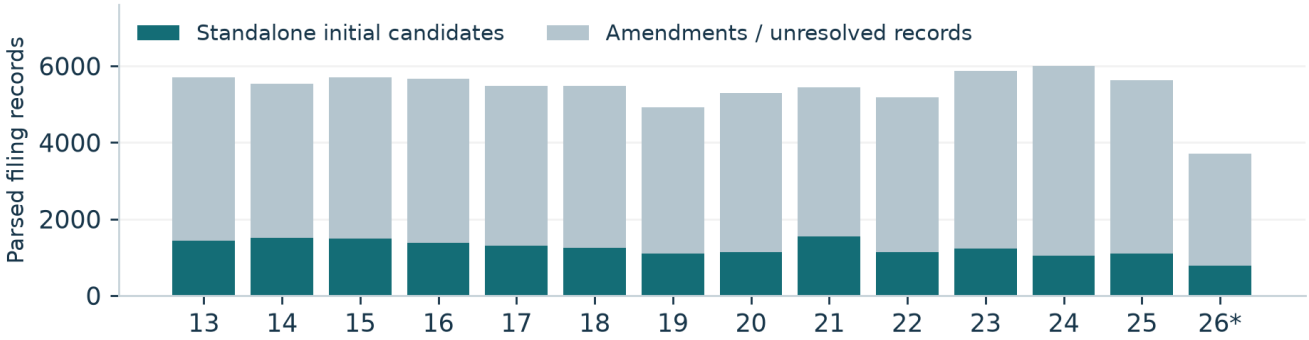
HFR fund performance is different from target-stock returns available to an outside follower. Dated manager identities, availability and incomplete dead-fund coverage need validation before the scorecards can become signals. [E3]

# From about 75,000 filings to 15,813 candidates

Cached SEC archive: January 2, 2013-September 11, 2026 | Unit: parsed filing records

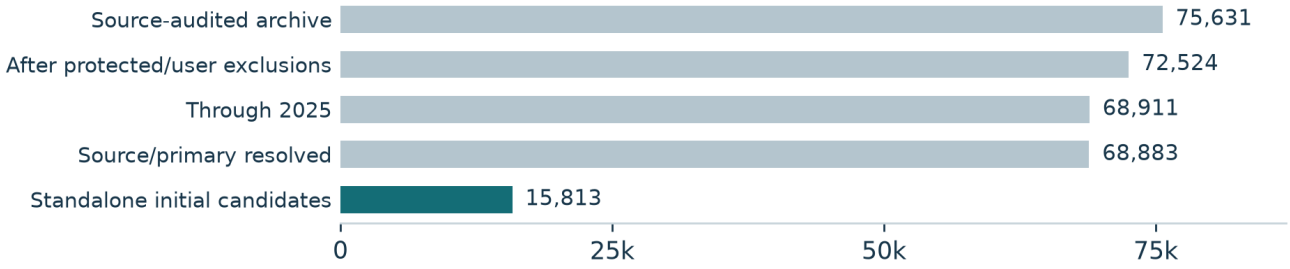
The audited archive contains **75,631 records**, including **58,166 amendments**, **17,420 standalone initial-filing candidates** and **45 unresolved records**. These are filing records, not unique companies, managers or independently validated campaigns. [E2]

## 3. Archive coverage by acceptance year




Years shown as 13-26. *2026 is partial through September 11. The initial-candidate segment is before the staging exclusions below. This is the project's cached, parsed archive, not a completeness claim about every SEC filing. [E2]

## 4. The preparation filter - counts reconcile in priority order




| Step / reason | Removed | Remaining |
|---|---|---|
| Source-audited archive | - | 75,631 |
| Protect review records/issuers; user-excluded Hill International | 3,107 | 72,524 |
| Restrict acceptance dates to December 31, 2025 or earlier | 3,613 | 68,911 |
| Remove unresolved source/primary-document cases | 28 | 68,883 |
| Retain standalone initial candidates; exclude remaining other records | 53,070 | **15,813** |

Each record is counted once under the first applicable exclusion. The final 53,070 includes remaining amendments and other non-initial records; it is not the full archive's amendment count. Staging rules do not screen for realized returns; Hill is a disclosed retrospective user exclusion. [E2]

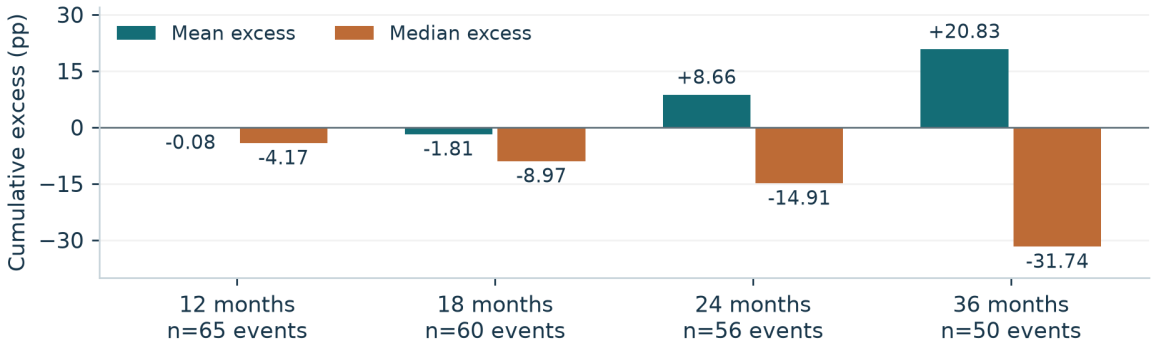
**What the 15,813 represent:** initial-filing candidates accepted in 2013-2025, all processed by the factual staging audit. They are not yet a validated hedge-fund universe or a tradable ML panel. Protected review samples remain independent; Hill's retrospective exclusion is a development choice.

**Separate return-study sample:** Phase 1 uses a 2018-2025 schedule and additional security, benchmark and outcome checks. At 24 months, the broad sample is 2,107 scheduled > 712 benchmark-assigned > 694 calculable; its focused activist-candidate subset is 67 > 58 > 56. The 56 events are not a direct funnel from the later 15,813-record staging exercise. [E1]

# Historical returns: positive averages, weak medians

Phase 1 exploratory development | 2018-2025 research window | Focused activist-candidate subset

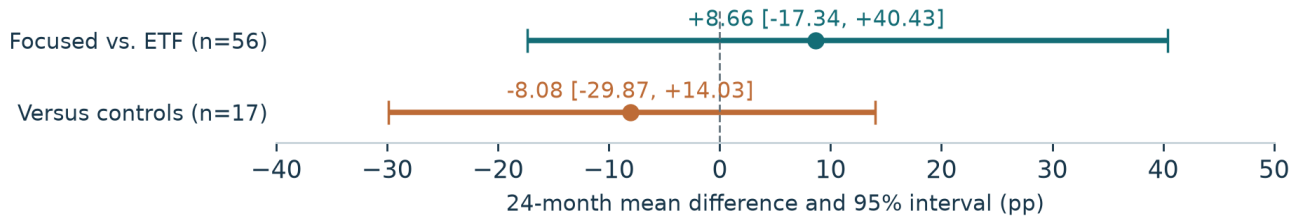
## 5. The mean improves with horizon; the median remains negative




| Hold | n | Stock mean | ETF mean | Excess mean | Excess median | Beat ETF | Mean excess 95% interval |
|---|---|---|---|---|---|---|---|
| 12 mo | 65 | +11.58% | +11.65% | -0.08 pp | -4.17 pp | 47.7% (31/65) | -11.25 to +13.64 pp |
| 18 mo | 60 | +13.52% | +15.34% | -1.81 pp | -8.97 pp | 35.0% (21/60) | -19.82 to +23.45 pp |
| 24 mo | 56 | +27.82% | +19.15% | +8.66 pp | -14.91 pp | 41.1% (23/56) | -17.34 to +40.43 pp |
| 36 mo | 50 | +53.57% | +32.74% | +20.83 pp | -31.74 pp | 32.0% (16/50) | -20.87 to +84.21 pp |

Gross, cumulative, equally weighted event returns; excess = stock minus its Russell size-segment ETF proxy; pp = percentage points. Horizons use different, overlapping calculable samples, so higher means do not establish an optimal holding period. Intervals use the saved issuer bootstrap. [E1]

## 6. Two-year results weaken under a matched comparison




Both 95% intervals cross zero. In 17 industry/year/size-matched cases, focused-event excess averaged -3.07 pp versus +5.01 pp for controls: difference -8.08 pp. This subset differs from the 56-event ETF comparison; the two estimates must not be pooled. [E1]

## 7. A funded account is a separate test

| Completed $100k / 12-month account | Result | Comparison / context |
|---|---|---|
| March 19, 2018-December 31, 2025 | +57.41% cumulative<br>+5.99% annualized | Cash-only +22.87%; SPY +181.73%; IWV +166.93% cumulative. |
| Deployment and downside | 75 buys; 55.41% average cash<br>-35.15% maximum drawdown | 5% issuer targets, max. 20; cash interest, liquidity limits and modeled costs. Different exposures prevent alpha claims. |

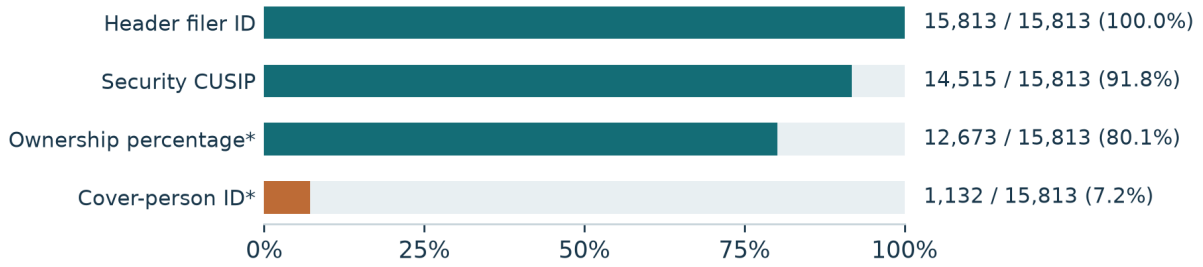
**Conventions:** modeled entry at the opening price on the second exchange session after SEC acceptance; 252/378/504/756 sessions approximate 12/18/24/36 months. Observed opens do not guarantee fills. No simultaneous duplicate-issuer positions in funded simulations. Historical stock/benchmark coverage is selective; monthly membership dates have availability limits.

Primary $100k 18/24/36-month accounts remain incomplete at Aratana's replacement-share/contingent-right settlement; proceeds are not invented. The gross event table does not resolve that account issue. Holding horizons were revised after inspecting earlier results; this is exploratory evidence. [E1]

# From prepared data to an investable selection test

Next research steps | Candidate mechanisms, not demonstrated alpha signals

## 8. Extraction coverage is substantial; economic links need work




Denominator: 15,813 selected 2013-2025 filings. *At least one extracted value per filing; this does not establish that every person or ownership row was recovered. Bars overlap. Header filer ID is not a validated fund identity; a CUSIP is not a verified historical stock link. Coverage measures presence, not accuracy. [E2]

## 9. Proposed signals and their investment mechanism

| Candidate signal | Economic hypothesis | Evidence needed |
|---|---|---|
| **Manager quality / history** Dated HFR peers; prior target-stock outcomes. | Repeated execution skill may distinguish managers whose plans create value. | Validate historical identities and availability; separate fund returns from outside-follower returns. |
| **Company economics** Valuation, leverage, cash flow, liquidity. | A credible catalyst may have more upside where value and financial capacity support it. | Build dated fundamentals/market features; test against size, industry and risk explanations. |
| **Disclosed intent (NLP)** Board change, asset sale, capital return; plan specificity. | A concrete, feasible plan may change cash flows or capital allocation beyond what prices reflect. | Validate text labels against source spans; compare ML with and without NLP on identical events. |
| **Stake and campaign updates** Ownership influence; later amendments. | Commitment and evolving engagement may improve conviction or exit decisions. | Resolve beneficial-owner overlap; test an amendment strategy separately from fixed holding periods. |

## 10. Validation gates before taking concentrated active risk

| Next step | What it must establish |
|---|---|
| **Make information historically available** | Resolve manager/stock links and missingness; use only facts public before purchase. Freeze targets, manager-quality rules and sample definitions. |
| **Test incremental selection value** | Use chronological evaluation and common samples/trading costs. Compare simple following, structured ML and ML plus NLP; report uncertainty, downside and coverage. |
| **Translate conviction into several positions** | Evaluate liquidity, loss scenarios, shared sector/factor/manager exposures and position sizes. Concentrated sizing is proposed; existing 5% targets are a baseline. |

Previously inspected history is development data. Reserve an unexposed or prospective evaluation under a frozen protocol before claiming a repeatable edge. The near-term milestone is a dated, validated feature panel and a credible selection comparison, followed by position-level risk evaluation.

**Evidence key:** E1: Sept 24 long-horizon summaries, matched controls and Phase 1 conclusion. E2: Sept 24 universe ledger/staging selection and Sept 25 coverage audit. E3: Sept 30 HFR protocol/scorecards. E4: frozen CRSP/FTSE source inventory, WRDS access records and early Yahoo preview script. Direct local source links in companion speaking notes.

Prepared October 4, 2026 from preserved project evidence. AI-assisted synthesis and graphics; no new backtest, trained model, intent labels or changes to independent human-review materials.